STATISTICAL STRUCTURED PREDICTION

Question set (Part 1-A)

November 2022

Administrative issues

- This quiz represents the fourth part of the final score of PEE.
- Each question has a score according to its difficulty.
- Answers can be handwritten, as long as this is readable.
- Please present the results in a "pdf" document and send it to me by email.
- Deadline for delivering these questions is **16 December 2022**

Question 1 (10 points)

Let be a Bayesian Network defined by the factorization associated with its joint probability.

$$P(C, D, H) = P(C)\ P(D \mid C)\ P(H \mid C)\,,$$

Where:

C	represents whether there are cavities (yes or no).
D	represents whether there is pain (yes or no).
H	represents whether there are holes (yes or no).

The probabilities are,

$P(C)$

si	0.3
no	0.7

$P(D \mid C)$

C	si	no
si	0.7	0.3
no	0.1	0.9

$P(H \mid C)$

C	si	no
si	0.9	0.1
no	0.2	0.8

What is the probability of having cavities if there are holds between the teeth and there is no pain?

Question 2 (10 points)

Given a HMM $(\mathcal{Q}, \Sigma, \theta)$ where: $\mathcal{Q} = \{1, 2, 3\}$ is the set of states in the model; $\Sigma = \{a, b\}$ is the set of output symbols; and θ is the vector of parameters that contains two types of parameters:

- $q(u|v): \; u, v \in \mathcal{Q}$ is the *transition probability* from state v to state u.

	$u=1$	$u=2$	$u=3$	$u=\#$
$v=1$	0.4	0.4	0.2	0.0
$v=2$	0.0	0.5	0.3	0.2
$v=3$	0.0	0.0	0.6	0.4
$v=\#$	0.8	0.2	0.0	0.0

  Note that $q(u|\#)$ represents the *initial state probability* of state, u, and $q(\#|v)$, represents the *final state probability* associated with state v.

- $e(o|u): \; u \in \mathcal{Q}, \; o \in \Sigma$ is the *probability of emitting symbol* o from state u.

	$o=a$	$o=b$
$u=1$	0.2	0.8
$u=2$	0.8	0.2
$u=3$	0.5	0.5

Given the grammar,

S → a X	X → a X	Y → a Y	Z → a Z
S → b X	X → b X	Y → b Y	Z → b Z
S → a Y	X → a Y	Y → a Z	Z → ϵ
S → b Y	X → b Y	Y → b Z	
	X → a Z	Y → ϵ	
	X → b Z		

The goal of this question is to complete the probabilities of rules so that this PCFG is equivalent to the HMM defined above. The PCFG will be equivalent to this HMM if for any sequence of input symbols, x, and their corresponding sequence of states, y, with probability, $P_{\text{HMM}}(x, y) > 0$, given by this HMM, it is fulfilled that:

1. There should be a parse tree, t, such that the pair, (x, t), gets probability, $P_{\text{PCFG}}(x, t) > 0$, given by the PCFG;
2. This probability must satisfy $P_{\text{HMM}}(x, y) = P_{\text{PCFG}}(x, t)$.

Prove it for a couple of derivations from the input sequence, $a \; b \; a$.

Question 3 (20 points)

Consider a bigram HMM, as introduced in class. We saw that HMMs could be defined in terms of transitions (factors), such as:

$$\Psi_t\,(y_{t-1},\; y_t,\; x_t) \;=\; q(y_t \mid y_{t-1}) \;\cdot\; e(x_t \mid y_t)$$

We also saw that the *Viterbi* algorithm for bigram HMMs could be used to calculate the probability of the best sequence of states of an observation. For this, we define the maximum probability that state $y_t = s \in \mathcal{Y}$ is reached at time t by emitting the prefix $x_1 \ldots x_t$ as:

$$\gamma_t(s) \;\stackrel{\text{def}}{=}\; \max_{y_1^t;\, y_t = s} \prod_{i=1}^{t} \Psi_i\,(y_{i-1},\; y_i,\; x_i) \;=\; \max_{y_1^t;\, y_t = s} \prod_{i=1}^{t} q(y_t \mid y_{t-1}) \;\cdot\; e(x_t \mid y_t).$$

In this question, we will consider a *"trigram"* model for HMMs, where the transition $\Psi_t(\,\cdot\,)$ takes the form,

$$\Psi_t\,(y_{t-2}, y_{t-1},\; y_t,\; x_t) \;=\; q(y_t \mid y_{t-2}, y_{t-1}) \;\cdot\; e(x_t \mid y_t)$$

We have assumed that $y_{-1} = y_0 = \#$ for $t = 1$. We call it a trigram HMM because we now consider sequences of three output labels (y_{t-2}, y_{t-1}, y_t).

Taking as reference the Viterbi algorithm seen in class, give a new version of this Viterbi algorithm for a trigram HMMs.

Question 4 (20 points)

As introduced in class, a CRF can be viewed as a factor graph where the bigram transition scores can be expressed as:

$$\Psi_t\,(y_{t-1},\; y_t,\; x_t) = \exp\,\{\sum_{k=1}^{K}\;\theta_k\; f_k\,(y_{t-1},\; y_t,\; x_t)\,\}$$

We also saw that the Forward algorithm for CRFs could be used to calculate the score of an observation,

$$\alpha_t(s) \quad \stackrel{\text{def}}{=} \quad \sum_{y_1^t;\; y_t = s} \;\; \prod_{i=1}^{t} \; \Psi_i\,(y_{i-1},\; y_i,\; x_i)$$

In this question, we will consider a *"skip"* model for CRF, where the transition, $\Psi_t(\,\cdot\,)$, takes the form

$$\Psi_t\,(y_{t-2},\; y_t,\; x_t) = \exp\,\{\sum_{k=1}^{K}\;\theta_k\; f_k\,(y_{t-2},\; y_t,\; x_t)\,\}$$

$\Psi_t(\,\cdot\,)$ is again a local transition score definition. We have assumed in this definition that $y_{-1} = y_0 = \#$. We call it a skip CRF because y_{t-1} is now omitted from the information that conditions the calculation of y_t.

Taking as reference the Forward algorithm seen in class, give a new version of this Forward algorithm for a *skip* CRF.

Question 5 (30 points)

Given context-free grammar in normal Chomsky form and an input sequence, x. The objective of this question is to design an algorithm that obtains the number of analysis trees for x. For example, if `a a a` is the input sequence and the context-free grammar is:

$$\text{S} \rightarrow \text{S S} \qquad \text{S} \rightarrow \text{a}$$

with start symbol, S, the algorithm should return the value 2 since there are only two parse trees for this sequence:

Suggestion: You can define a score $\pi(A, i, i+l)$ that represents the number of possible parse trees that have A as the solution of subproblem $x_{i+1} \ldots x_{i+l}$.

Question 6 (30 points)

Given the CKY-based *Viterbi* algorithm for PCFGs seen in class, this algorithm calculates the probability of the best parse tree of an input sequence $x_1, x_2, \ldots, x_T$. Where the initialization step is, $\forall A \in N$ and $\forall i = 0 \ldots T-1$,

$$\widehat{e}(A < i, i+1 >) \;=\; p(A \rightarrow b)\; \delta(b, x_{i+1})\,.$$

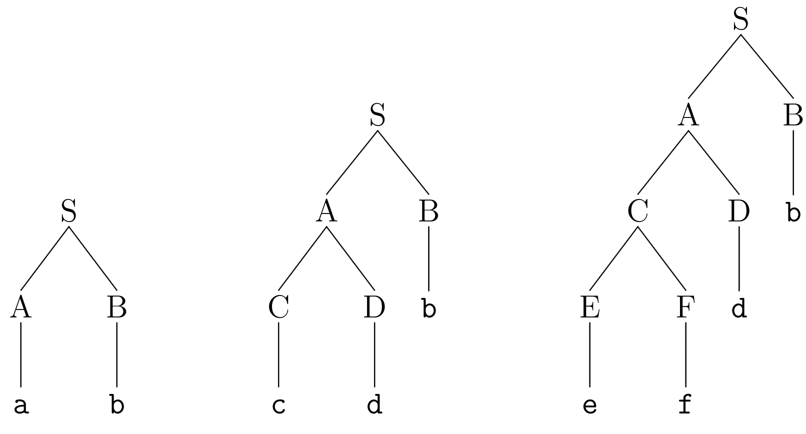
Moreover, the recursion step is as follows: $\forall A \in N$ and $l : 2 \ldots T$, $\forall i : 0 \ldots T-l$,

$$\widehat{e}(A < i, i{+}l >) \;=\; \max_{B,C \in N} p(A \rightarrow BC) \max_{k=1,\ldots,l-1} \widehat{e}(B < i, i{+}k >)\, \widehat{e}(C < i{+}k, i{+}l >)\,.$$

Finally, the probability of the best parse tree is:

$$\widehat{P}_{G_\theta}(x) = \widehat{e}(S < 0, n >)$$

Now assume that we want to compute the probability of the best parse tree of an input sequence for a *l*eft-branching grammar. Here are some example left-branching trees:



As can be seen in these examples, the left-branching parse trees involving binary rules of form $X \rightarrow YZ$, the non-terminal Z must be derived directly by a terminal symbol.

Give a new version of the CKY-based Viterbi algorithm to calculate the probability of the best parse tree of an input sequence, $x_1, \ldots, x_T$, for (*left-branching*) grammars. In addition, the temporary cost of the algorithm proposed must also be determined.